
Introduction to Human Physiology

Physiological Adjustment

Homeostasis

- The tendency of biological systems to maintain a state of equilibrium
- A process that maintains the internal environment of living systems in a more or less constant state
- May be occasionally interrupted when a stressor is perceived and a **fight-flight response** is in effect
- **Measured values** are compared with a **pre-determined "set point"**
- **Error signal** → the difference between measured values and **pre-determined set point**
 - The error signal is used to generate signals that help return the regulated variable toward its pre determined value

Homeostasis

- Components of Homeostasis
 - Set point → predetermined reference value
 - Receptor → detects changes
 - Afferent path → transmits signal from the receptor to an integrating center through afferent nerves or hormones
 - Integrating center → assess and discriminate the transmitted signal using a reference or set value
 - Efferent path → transmit the discrepancy (error signal) that occurs to effector organs through efferent nerves
 - Effector organ → respond in accordance with the need for adjustment

Homeostasis

- Regulation
 - Both intrinsic and extrinsic
- Intrinsic regulation:
 - Occurs in cell/tissue/organ
 - These body components adjust themselves intrinsically to environmental changes
 - They are also adjusted by extrinsic regulation
- Extrinsic regulation
 - Involves **nervous + endocrine systems** that constitute long distance communication systems

Homeostasis

- Extrinsic regulation
 - Nervous system → Rapid adjustment to changes
 - Endocrine system → slower but long lasting metabolic adjustment
 - Neuro-endocrine → integrated in many instances to provide optimal environment for the cell:
 - Neurohypophyseal system → hypothalamus – pituitary gland
 - Autonomic – endocrine system → SNS – Adrenal gland

Homeostasis

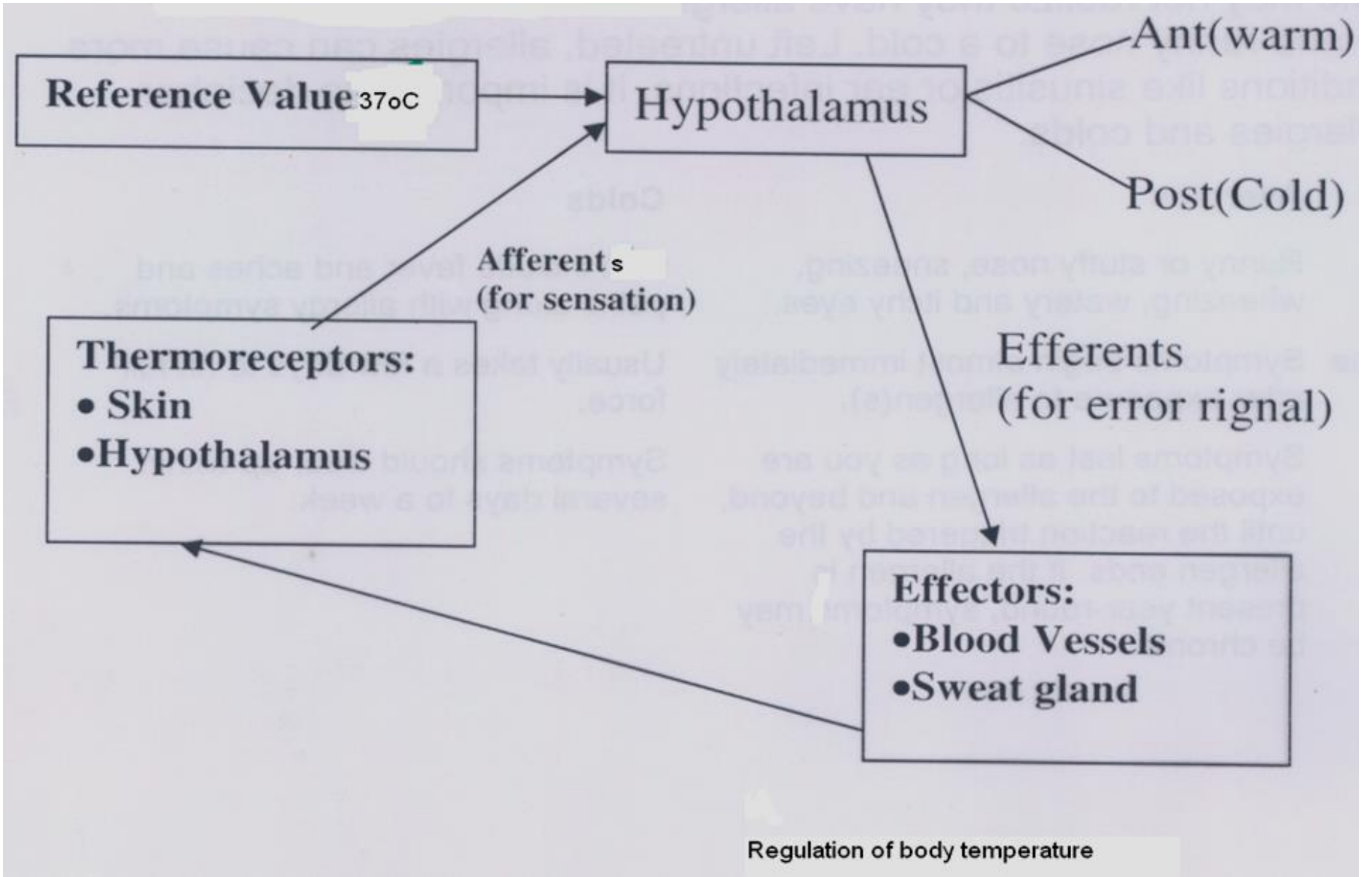
- Extrinsic regulation: components
 - Receptors
 - Nerves
 - Hormones
 - Coordinating centers
 - Enzymes and Nutrients
 - Effectors

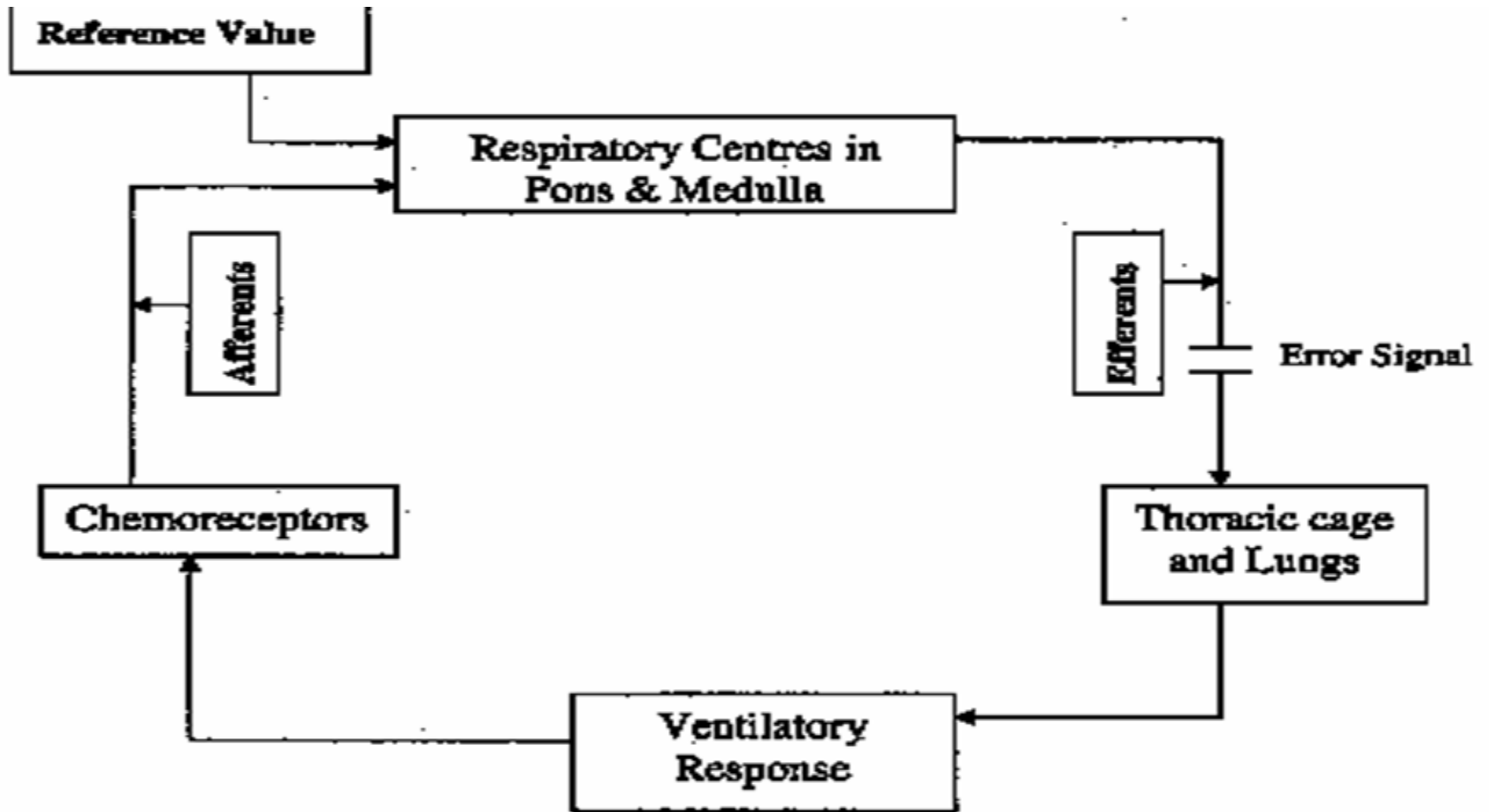
Homeostasis

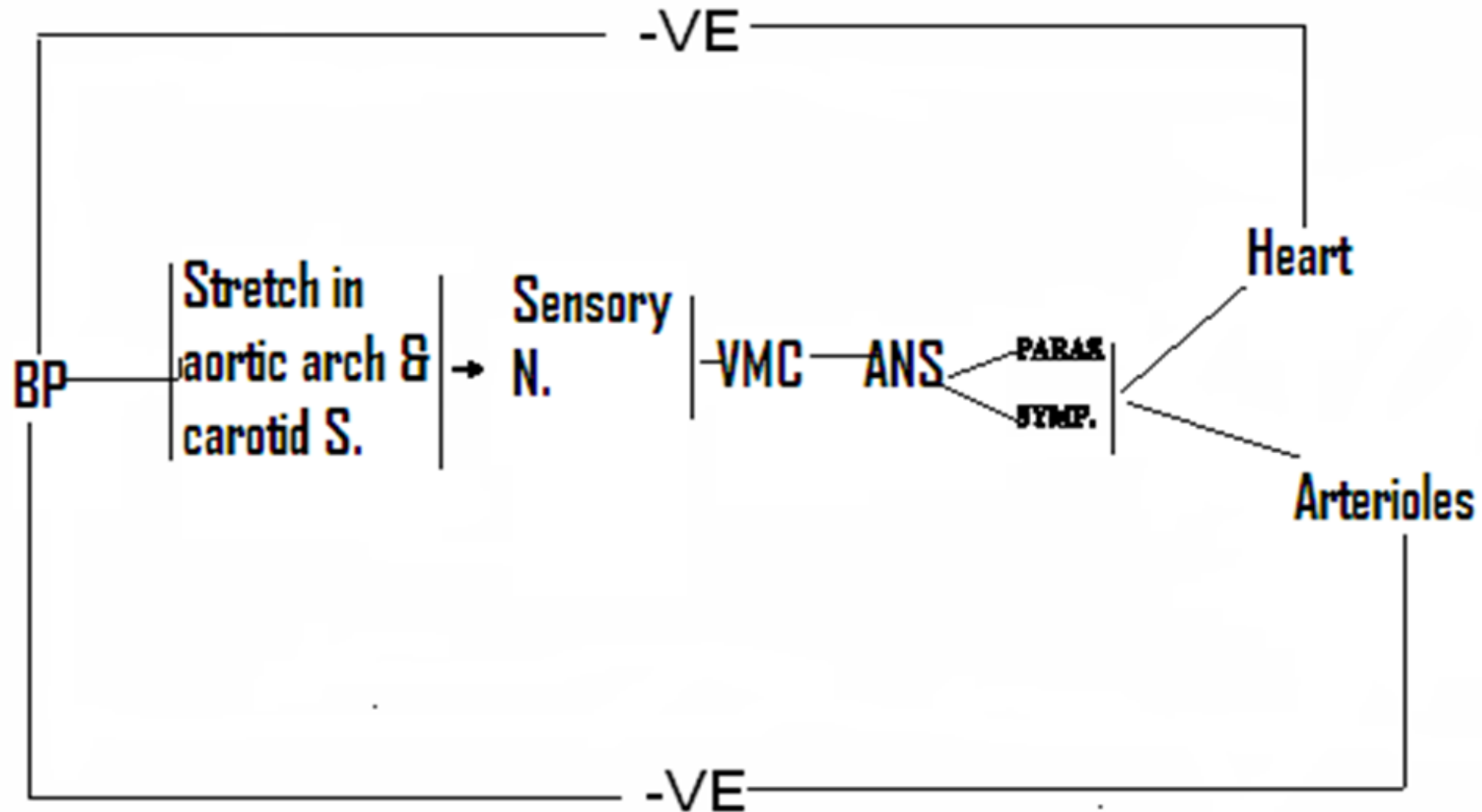
- Negative Feedback
 - Output returns to the normal input → response is in the opposite direction as the stimulus
 - The system is restored back to its original state by resisting or opposing any attempts to change the system → “The law of sheer cussedness” favoring stability

Homeostasis

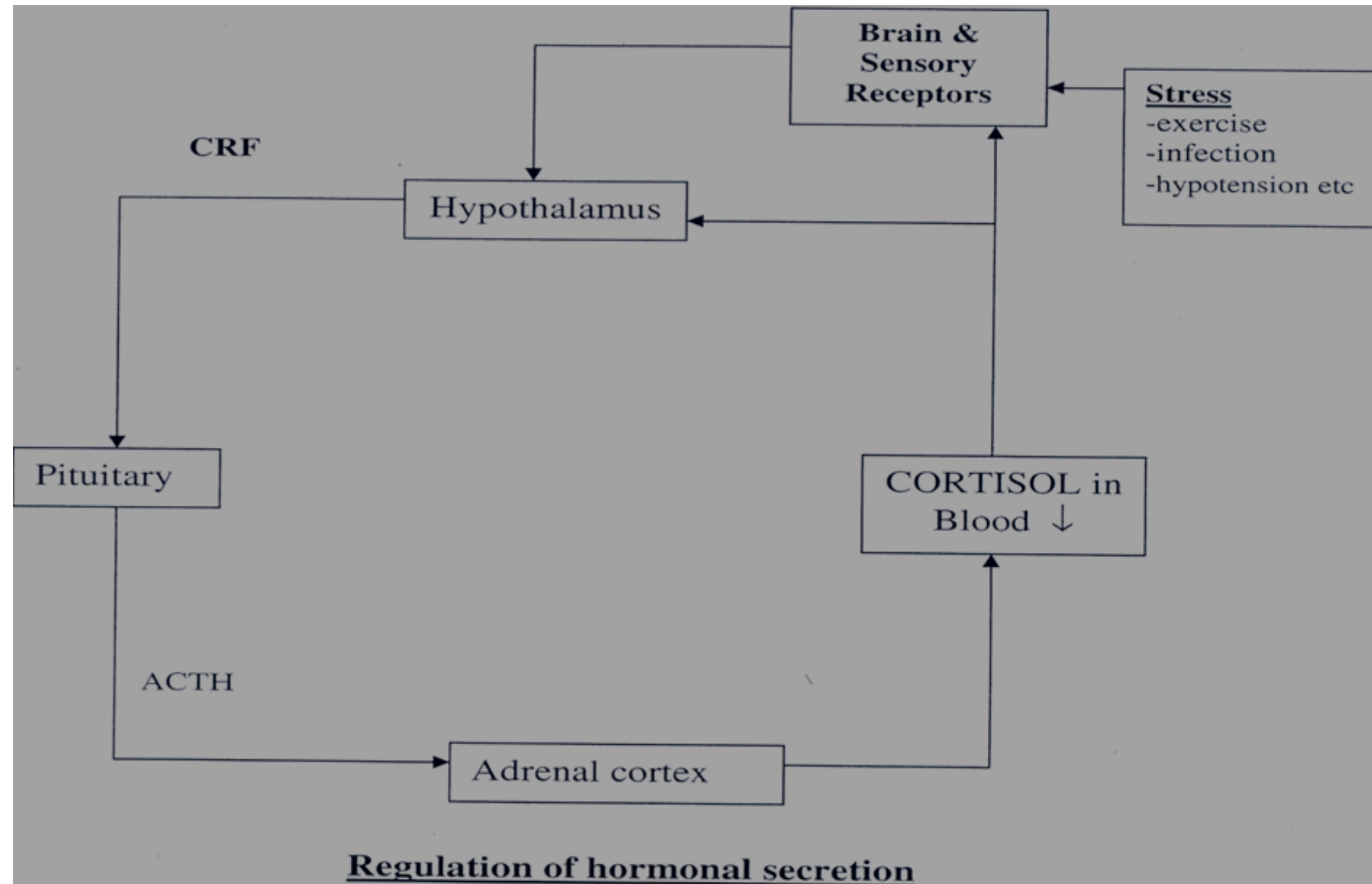
- Negative Feedback
 - Regulation of Temperature
 - Regulation of PaCO₂
 - Regulation of Blood pressure
 - Regulation of hormonal secretion
 - Regulation of blood glucose
 - Regulation of effective circulating volume

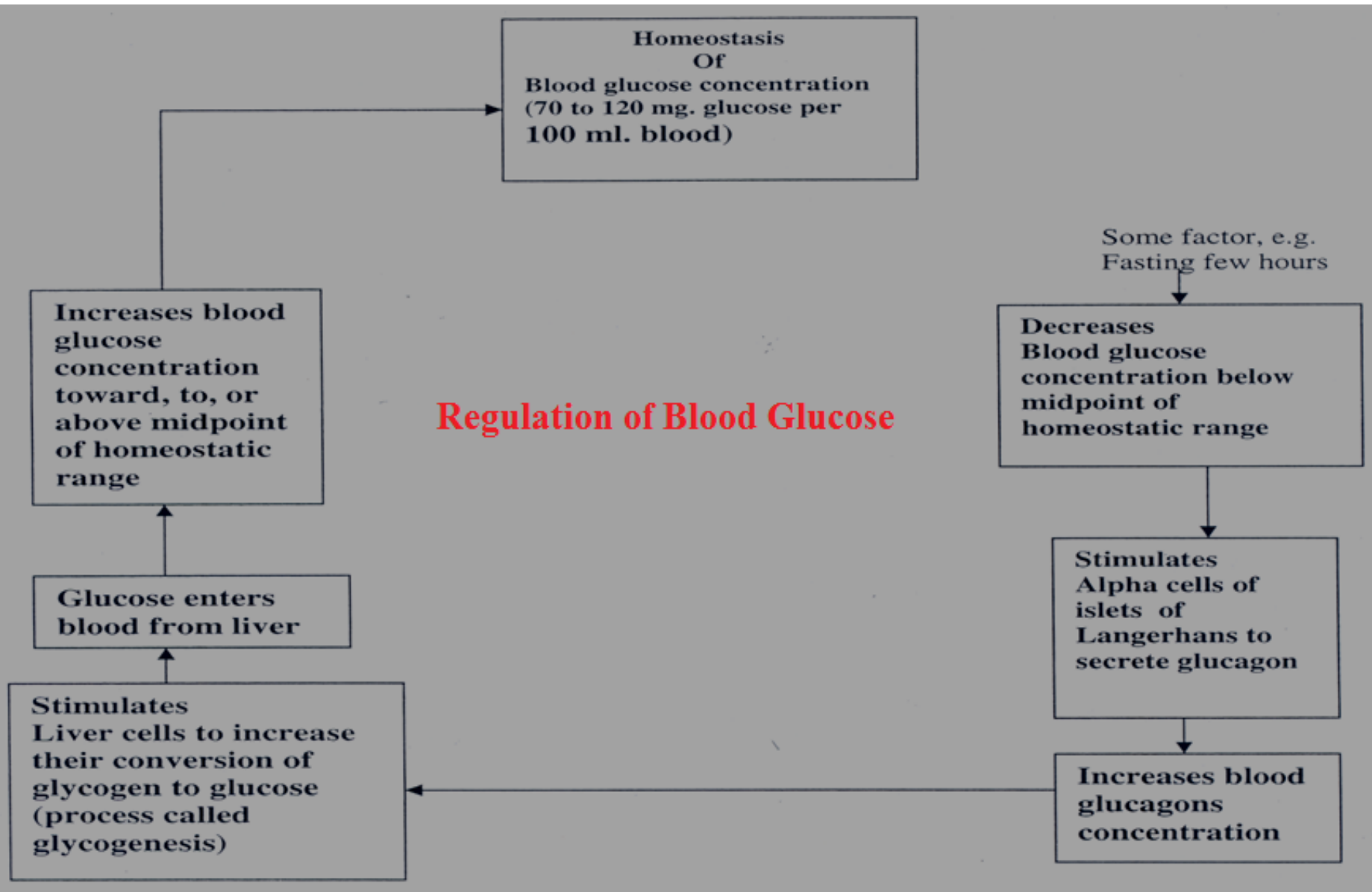


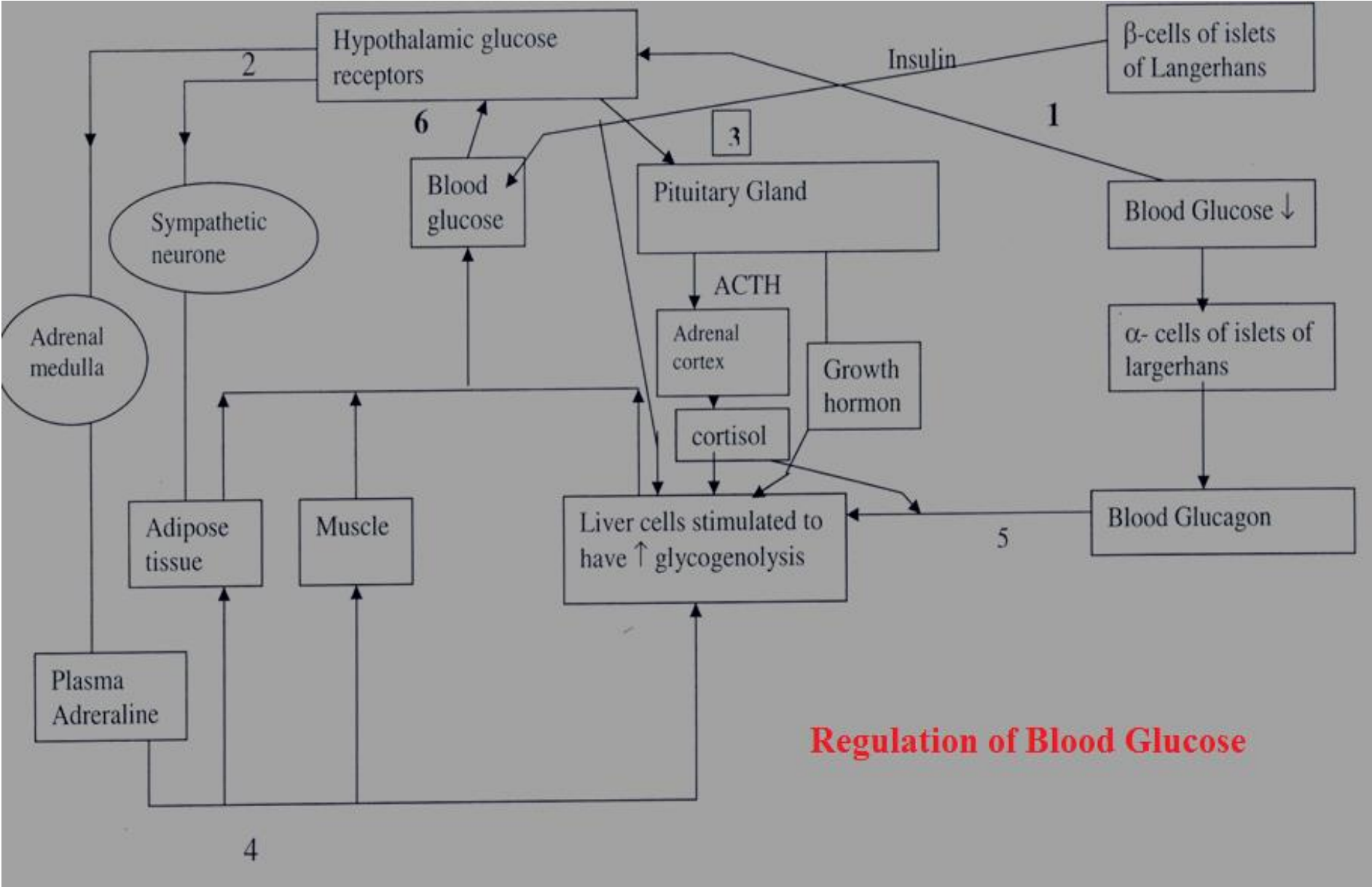
Regulation of PaCO_2



Regulation of BP







Homeostasis

- Regulation of Effective Circulating Volume (ECV)

1. Responses to acute hemorrhage ($\downarrow$ ECV)

- Acute hemorrhage evokes the following negative feedback mechanisms

- i. Hemostatic adjustment

- ii. Baroreceptor reflexes

- Peripheral systemic vasoconstriction
- $\uparrow$ HR + myocardial contractility $\rightarrow$ $\uparrow$ SV $\rightarrow$ $\uparrow$ BP
- $\downarrow$ RBF

Regulation of ECV

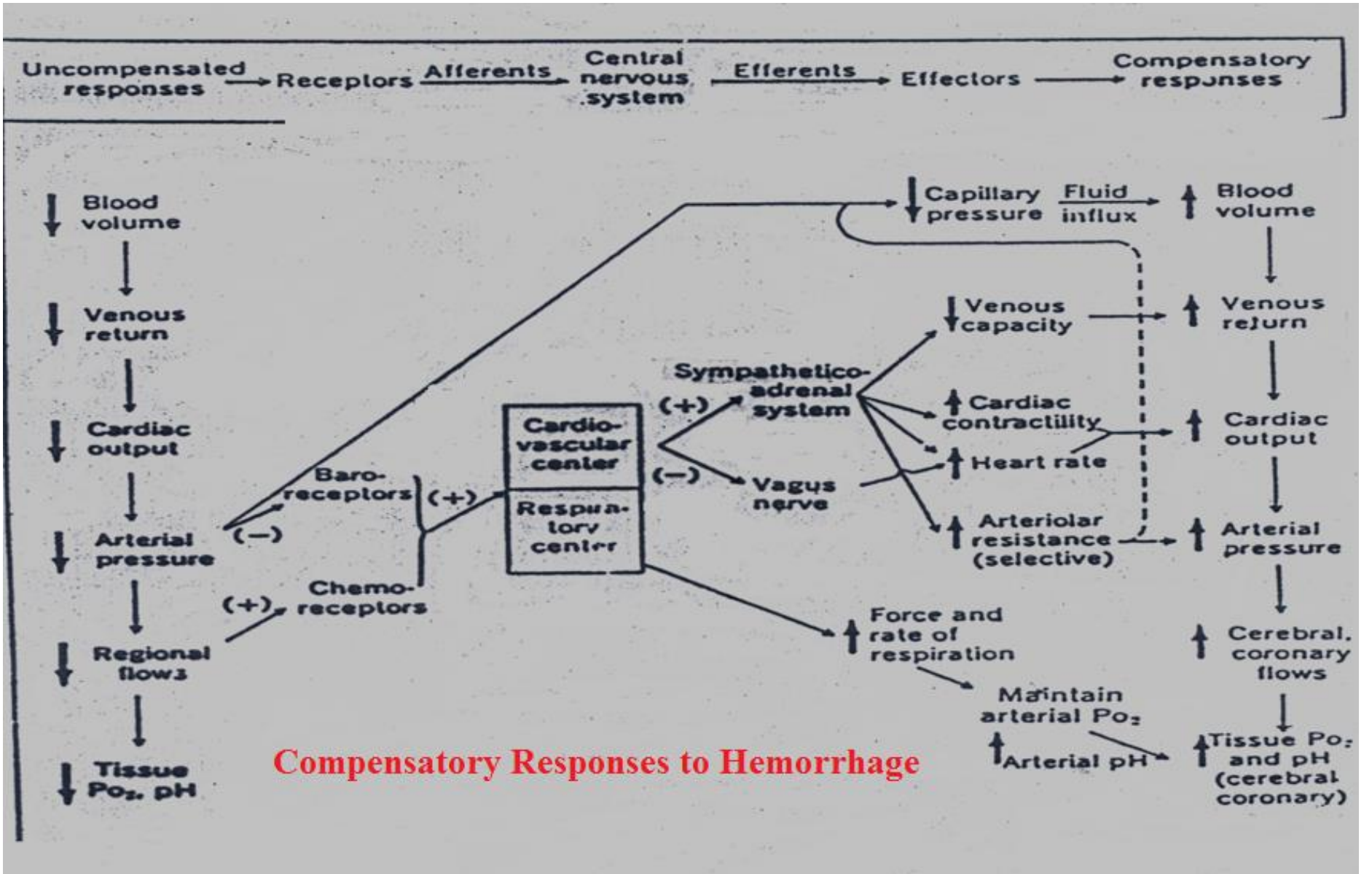
iii. Chemoreceptor reflexes:

- $ABP < 60\text{mmHg}$ → Inadequate blood flow to aortic and carotid bodies → reduced PO_2 in chemoreceptor tissues → augmentation of vasoconstriction previously induced by baroreceptor mechanism

iv. Cerebral ischemic responses:

- $ABP < 40\text{mmHg}$ → Activate SNS → intense sympathetic discharge → pronounced CVS response

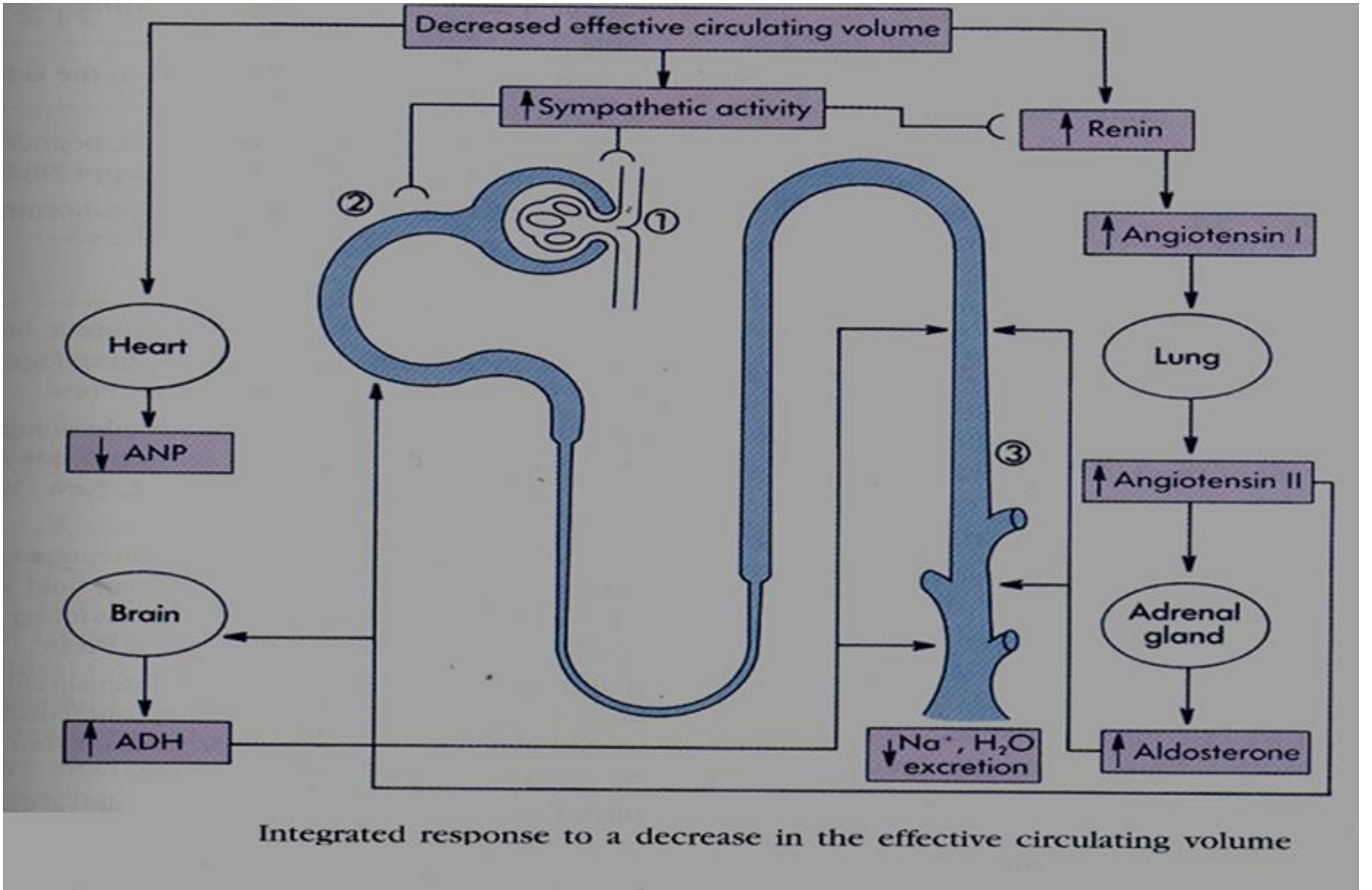
v. ↓Capillary Pressure → shift of fluid from IF to intravascular compartment

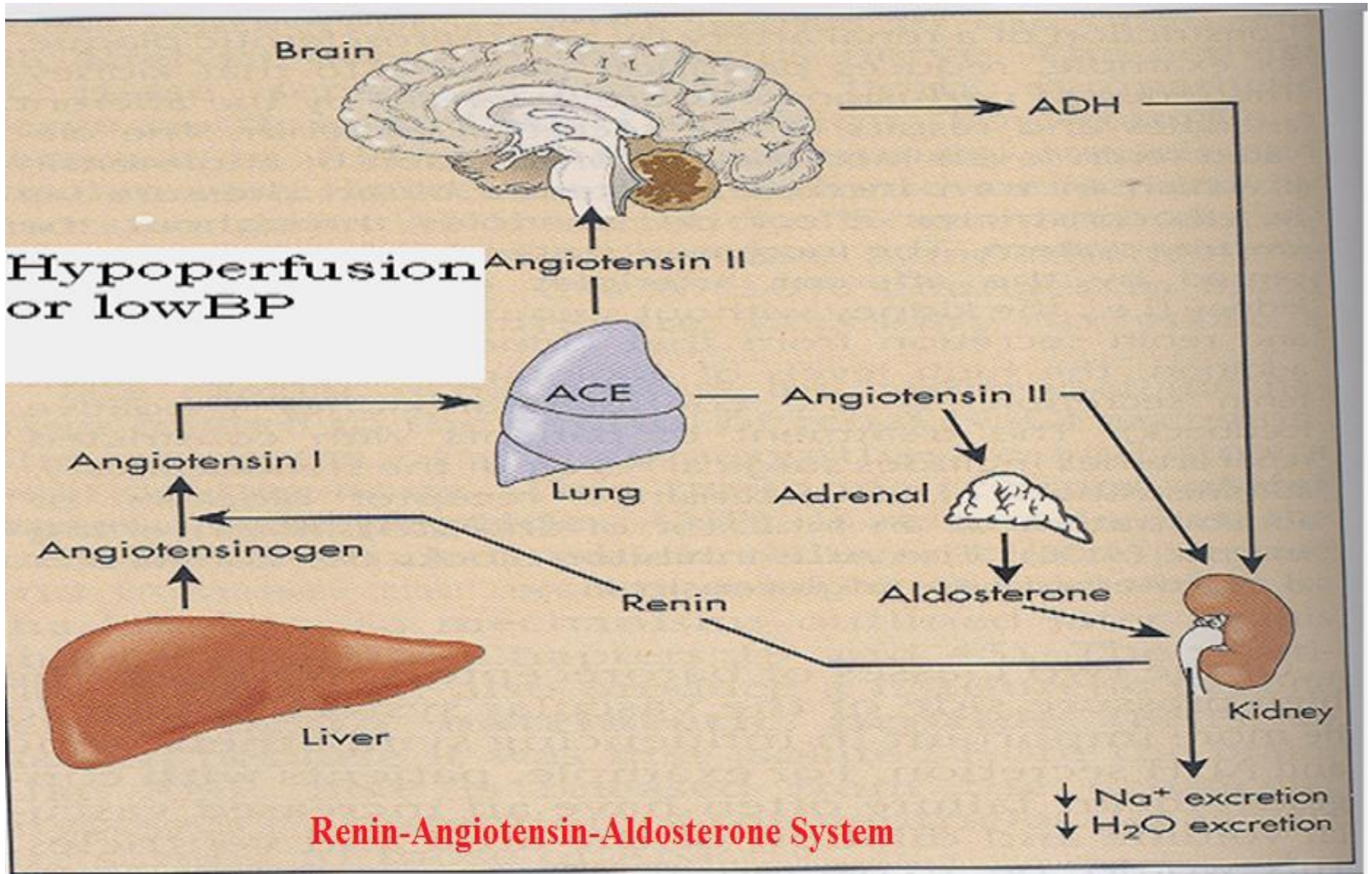


Regulation of ECV

vi. Renal conservation of salt and water

- Reduced GFR → reduced excretion of water and electrolytes:
 - a. **GFR and filtered load of Na^+ is reduced** – due to afferent and efferent arteriolar constriction mediated by the SNS
 - b. **Na^+ reabsorption** by the proximal tubule is stimulated due to increased sympathetic activity and the direct action of Ang-II
 - c. **Na^+ reabsorption** through the collecting duct is enhanced by the action of aldosterone and ADH (ANP increases excretion)
 - d. **Renin-Angiotensin-Aldosterone System** → retention of Na^+ and water





Regulation of ECV

vii. Release of endogenous vasoconstrictor substances/hormones:

- $\uparrow$ Catecholamines $\rightarrow$ Vasoconstriction + $\uparrow$ cardiac contractility
- $\uparrow$ Glucagon + Cortisol $\rightarrow$ $\uparrow$ BS $\rightarrow$ $\uparrow$ tonicity $\rightarrow$ $\uparrow$ ADH $\rightarrow$ $\uparrow$ Vasomotion $\rightarrow$ $\uparrow$ Volume through “atrial volume reflex (Gauer-Henry reflex)”
 - **ADH** $\rightarrow$ potent vasoconstrictor released into blood during acute haemorrhage ($\approx$ 20% blood loss leads to 40 fold increase in ADH secretion)
- $\uparrow$ Erythropoietin

Regulation of ECV

- The effects of haemorrhage may be decompensated by **positive feedback mechanisms**
- When haemorrhage is severe, positive feedback mechanism may be evoked
- This aggravates the primary changes initiated by blood loss and may initiate vicious cycles that are fatal

Circulatory Shock

- A failure of the circulation with decreased perfusion and/or inadequate oxygenation of the vital organs and cells of the body leads to circulatory shock
- Stages of shock → **Preshock → Shock → End Organ Dysfunction**

Circulatory Shock

- Pre shock
 - Also referred as warm shock or compensated shock
 - Characterized by rapid compensation for diminished tissue perfusion by various homeostatic mechanisms
 - Tachycardia, peripheral vasoconstriction, and either a modest increase or decrease in systemic blood pressure may be the only clinical signs of shock

Circulatory Shock

- Shock (decompensated):
 - During shock, the compensatory mechanisms become overwhelmed and signs and symptoms of organ dysfunction appear
 - These include tachycardia, dyspnea, restlessness, diaphoresis, metabolic acidosis, oliguria, and cool clammy skin...

Circulatory Shock

- Shock (decompensated):
 - Decreased perfusion of cells $\rightarrow \uparrow [\text{Na}^+]_i$ + while K^+ leak out
 - Hypo perfusion state $\rightarrow$ hypoxia $\rightarrow$ inability of mitochondria to produce energy $\rightarrow$ cell membranes become damaged $\rightarrow$ leaking in of ECF
 - Anaerobic metabolism continues $\rightarrow$ increased metabolic acidosis $\rightarrow$ arteriolar smooth muscle and pre-capillary sphincters relax thus helping blood to remain in the capillaries $\rightarrow \uparrow$ hydrostatic pressure combined with histamine release $\rightarrow$ leakage of fluid and protein into the surrounding tissues

Circulatory Shock

- End-organ dysfunction (Refractory) stage:
 - Progressive end-organ dysfunction leads to irreversible organ damage and patient death
 - During this stage, urine output may decline further (culminating in anuria and acute renal failure)
 - Acidemia decreases the cardiac output and alters cellular metabolic processes, and restlessness evolves into agitation, obtundation, and coma
 - Brain damage → cell death → death of organism

Circulatory Shock

- Stages of Recovery
 - Fluid volume.....24hrs
 - Plasma proteins....2-4days
 - RBC and Hba few weeks

Regulation of ECV

2. Responses to an increase in ECV

- Body fluid volume and osmolality are maintained within a narrow range with the kidneys playing the central role

i. GFR increases

- $\downarrow$ SNS activity across afferent arterioles $\rightarrow$ VD $\rightarrow$ $\uparrow$ glomerular capillary pressure $\rightarrow$ $\uparrow$ GFR
- This response is augmented by ANP through dilatation of both afferent and efferent arterioles
- Filtered load of Na^+ increases in parallel with GFR

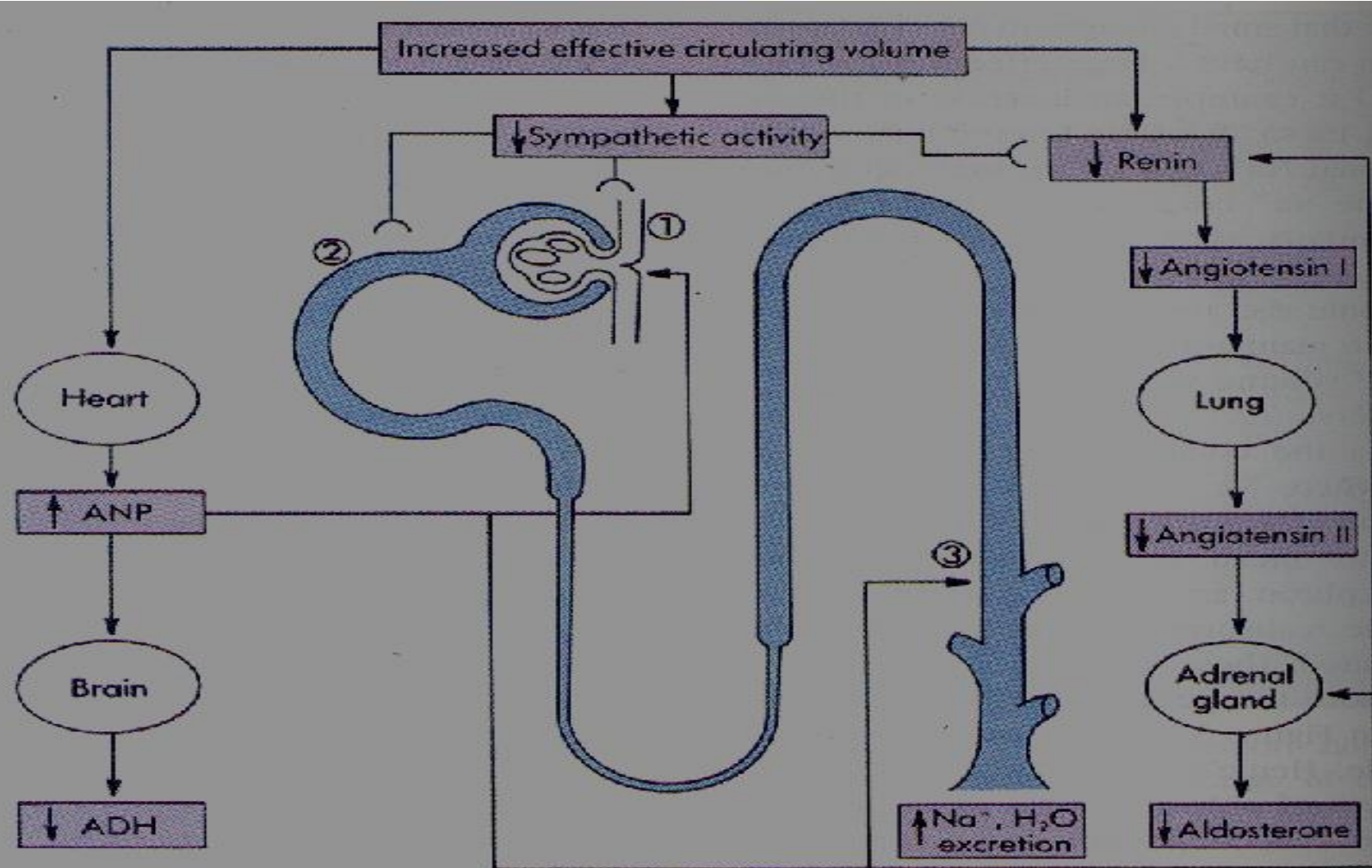


Fig-- Integrated response to an increase in the effective circulating volume. The numbers along the nephron indicate sites where sodium handling is regulated.

Regulation of ECV

- ii. Na^+ reabsorption decreases in the proximal tubule in response to reduced Ang-II
- iii. Na^+ reabsorption decreases in the collecting duct
 - Increase in the filtered load + decrease in proximal tubule reabsorption enhance the delivery of Na^+ to the loop of Henle, which in turn increases the delivery of Na^+ to the collecting duct

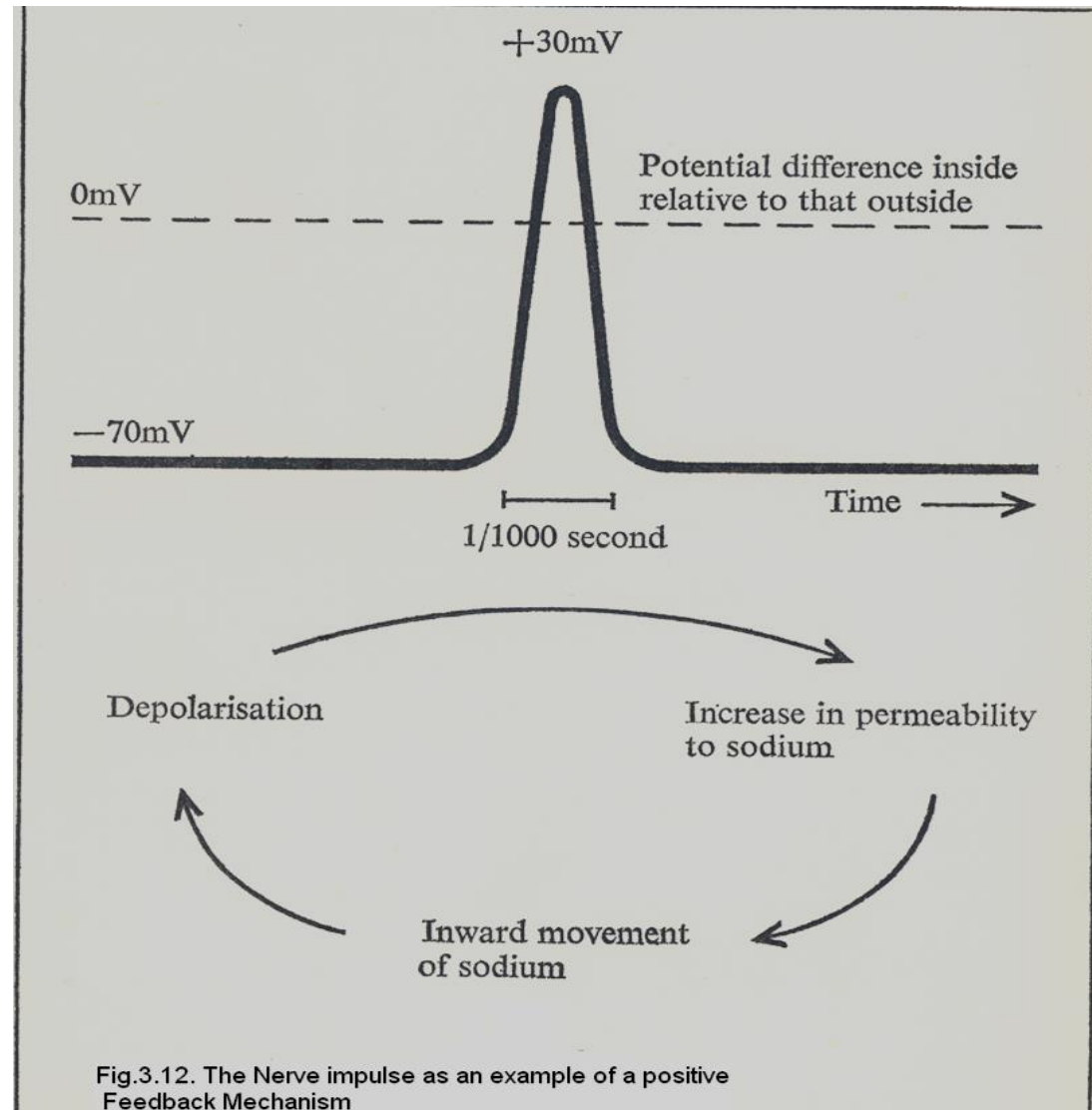
Homeostasis

- Positive Feedback
 - Output is not returned to the normal input:
 - i.e. response is in the same direction as the stimulus
 - May be part of a negative feedback mechanism
 - Rare in biology because in many cases, it causes instability leading to death, i.e. it favors instability

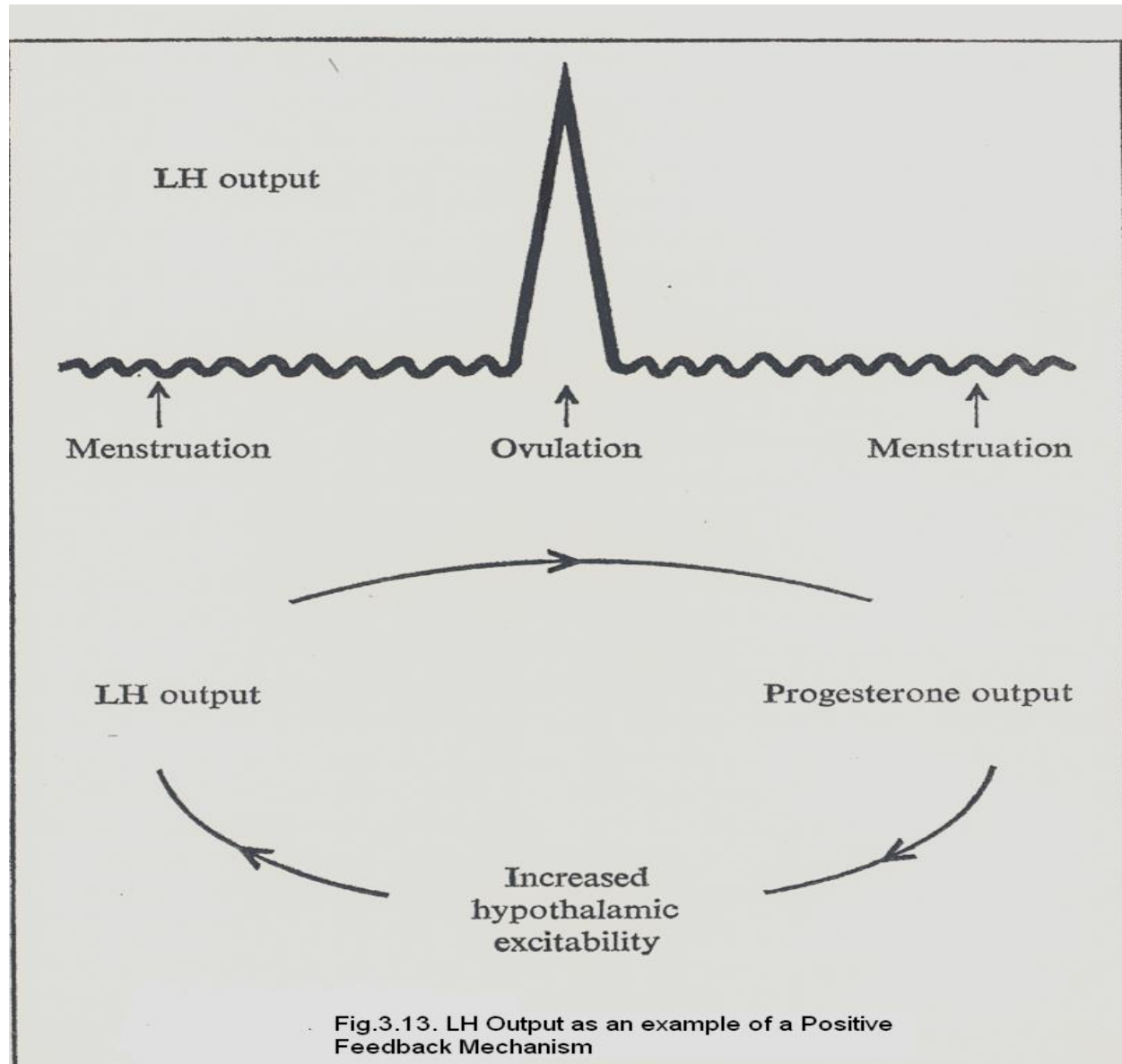
Homeostasis

- Positive Feedback
 - Includes:
 - Nerve impulse (AP)
 - LH secretory surge
 - Uterine contraction during labor
 - Severe (decompensated) shock
 - Viral infection: virus invades cell → replicates in cell → many viruses → invasion of other cells → further replication

- The nerve impulse:
 - Depolarization with progressively increasing Na^+ influx
 - Repolarization with progressively increasing efflux of K^+



- LH secretory surge in mid-cycle



Allostasis

- Maintenance of **stability and viability through change in the face of stress challenge** to represent acute stress response
- Highlights our ability to anticipate, adapt to, or cope with impending future events
- **Allostasis:** maintaining stability (or homeostasis) through acute change, i.e. attaining viability through change
- It addresses basic regulatory systems and examines the behavior of bodily regulation under duress

Allostasis

- With allostatic regulation by cephalic involvement, the body adapts to potentially diverse and dangerous situations through the activation of neural, hormonal, or immunological mechanisms
- Various types of stress mediators such as catecholamines and cortisol are used in the process of Allostasis

Allostasis Load

- Allostatic load refers to the cost of adaptation to a stressful environment eliciting repeated and sometimes prolonged adaptive responses
- Repeated cycles of allostasis and inefficient turning- on or shutting off of responses lead to wear and tear (allostatic load) of the body
- Manifestations of allostatic load include:
 - Decreased cognitive function during aging,
 - Abdominal obesity, an increased risk for hypertension and cardiovascular disease,
 - Insulin-dependent diabetes, and
 - Decreased immune responses, etc.

Allostasis Load

- The concept of allostasis and allostatic load envisions a cascade of cause and effect that begins with primary stress mediators such as catecholamines
 - i. **Overstimulation by frequent stress**, resulting in excessive stress hormone exposure
 - ii. **Failure to inhibit allostatic responses** when they are not needed or an inability to habituate to the same stressor, both of which result in overexposure to stress hormones; and

Allostasis Load

- iii. Inability to stimulate allostatic responses when needed, in which case other systems (e.g., inflammatory cytokines) become hyperactive and produce other types of wear and tear

Adaptation (Heterostasis)

- Adjusting by changing chemically and structurally on a long-term basis
- External world → problems to be solved → structural evolutionary changes to solve problems of existence → adaptation

1. Long-term adaptation to cold

- 25-50% ↓BMR → cooling of peripheral and deep lying tissues → ↓average skin temp. up to 15-12 °C → gradient of heat flow reduced
- Thermal conductivity drops from 8 to 5 kcal/m²/hr/°C → ↓temp. gradient → reduced heat loss

Heterostasis

2. Long term Adaptation to heat

- Sweat glands:
 - Temperate climate → some sweat glands are inactive → reduced ability of sweating
 - Hot climate → all sweat glands become active → increased ability of sweating
- Genetic adaptation
 - Body shape → Long limbs

Heterostasis

3. Adaptation to high altitude

- Hyperhemoglobinemia
- Hypervascularization
- Hyperactivity of cell enzymes which $\uparrow$ O₂ utilization

Heterostasis

3. Adaptation to high altitude

- **Birds** are more adaptable to high altitudes because of **high cardiac output and more efficient respiratory system**
- **Mammals** are rarely found at about 6000 metres above sea level but some birds nest at 6500m and others migrate above 9000m
- The physiological adjustment is in **P50 and not in Hct**
- There are homeostatic and adaptive changes in organ systems including respiratory, cardiovascular/blood, endocrine and neural changes